

ASSIGNMENT UNIT 4

Conceptual and Analytical Problems

- Q1.** A code has minimum distance $d=4$.
Can it correct 2 errors? Justify mathematically. (BTL: AN)
- Q2.** Two codes have same length but different minimum distances:
Code A: $d=3$, Code B: $d=5$
Which is better and why? (BTL: EV)
- Q3.** A student claims:
"If a code detects 3 errors, it can also correct 3 errors."
Verify this statement. (BTL: AN)
- Q4.** For a $(7,4)$ Hamming code, a received word gives syndrome 000.
Is it always error-free? Explain edge case. (BTL: AN)
- Q5.** A code detects up to 2 errors. What is its minimum distance? (BTL: AP)

Multi-Step Analysis

Q6. A linear code has generator matrix:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Calculate:

- Encode message (1 0)
- Find all possible codewords
- Determine minimum distance (BTL: AN)

Q7. A received word 1101011 is obtained in a (7,4) Hamming code.

Calculate:

- Find syndrome
- Locate error bit
- Correct the codeword
- Identify original message (BTL: AN)

Q8. A cyclic code uses generator polynomial:

$$g(x) = x^3 + x + 1$$

Message: 101101

Calculate:

- Perform CRC encoding
- Introduce single-bit error and verify detection (BTL: AN)

Q9. A code has minimum distance 5.

Calculate:

- Maximum detectable errors
- Maximum correctable errors
- What happens if 3 errors occur? (BTL: AN)

Q10. Given parity check matrix, a student decodes incorrectly due to wrong syndrome mapping.

Calculate:

- Identify possible mistake
- Recompute correct decoding (BTL: EV)